

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

Gateway of last resort is not set

```

3.0.0.0/32 is subnetted, 1 subnets
C    3.3.3.3/32 is directly connected, Loopback0
O    192.168.1.0/24 [110/65] via 192.168.13.1, 00:13:46, Serial0/0/0
O    192.168.2.0/24 [110/129] via 192.168.13.1, 00:01:06, Serial0/0/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

```

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Neighbor ID	Pri	State	Dead Time	Address	Interface
11.11.11.11	0	FULL/ -	00:00:30	192.168.13.1	Serial0/0/0

R3#

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

All traffic from R3 to the 192.168.2.0/24 network will be routed via R1. The S0/0/1 interface on R2 remains configured as passive, so OSPF routing updates are not being advertised across that interface. The accumulated cost of 129 arises because packets from R3 to 192.168.2.0/24 traverse two T1 serial links (each with a cost of 64) plus the R2 Gigabit 0/0 LAN link (cost 1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

```

Router(config)#router ospf 1
Router(config-router)#no passive-interface s0/0/1

```

- i. Re-issue the **show ip route** command on R3.

```
192.168.1.0/24 [110/65] via 192.168.13.1, 00:23:35, Serial0/0/0
192.168.2.0/24 [110/65] via 192.168.23.1, 00:00:12, Serial0/0/1
```

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

R3 s0/0/1 192.168.2.0/24 64+1=65 The total cost is 65, which comes from one T1 serial link (cost 64) plus the R2 Gigabit 0/0 LAN link (cost 1).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Equipment available today commonly supports link speeds above 100 Mb/s. To calculate cost more accurately for these faster links, increase the default reference-bandwidth setting.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

```
O 192.168.23.0 [110/845] via 192.168.13.2, 00:00:05, Serial0/0/1
    [110/845] via 192.168.12.2, 00:00:05, Serial0/0/0
```

The cost to reach 192.168.3.0/24 is 782, R1 s0/0/1 192.168.23.0/30 781+64=845 The cost to reach 192.168.23.0/30 is 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

```
R1#show ip route ospf
O 192.168.2.0 [110/782] via 192.168.12.2, 00:09:16, Serial0/0/0
O 192.168.3.0 [110/782] via 192.168.13.2, 00:09:16, Serial0/0/1
192.168.23.0/30 is subnetted, 1 subnets
O 192.168.23.0 [110/1562] via 192.168.12.2, 00:01:07, Serial0/0/0
    [110/1562] via 192.168.13.2, 00:01:07, Serial0/0/1
```

The total cost is 1,562. Each serial link now has a cost of 781, and the path to 192.168.23.0/24 uses two serial links (781 + 781 = 1,562).

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

```
R1#show ip route ospf
O    192.168.2.0 [110/782] via 192.168.12.2, 00:13:10, Serial0/0/0
O    192.168.3.0 [110/1563] via 192.168.12.2, 00:00:06, Serial0/0/0
    192.168.23.0/30 is subnetted, 1 subnets
O    192.168.23.0 [110/1562] via 192.168.12.2, 00:00:06, Serial0/0/0
```

OSPF selects the path with the lowest total cost. The lowest-cost path is R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, with a cost of $781 + 781 + 1 = 1,563$. This is lower than the alternative path R1-S0/0/1 + R3-G0/0, which totals $1,565 + 1 = 1,566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router IDs affect which routers become the Designated Router (DR) and Backup Designated Router (BDR) on multi-access networks. If a router ID is tied to a physical interface, it might change when that interface goes down. To avoid this, use a loopback address (which stays up) or explicitly set the router ID with the router-id command.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR elections only occur on multi-access networks such as Ethernet or Frame Relay. The lab uses point-to-point serial links, so DR/BDR elections do not take place.

3. Why would you want to set an OSPF interface to passive?

Making a LAN interface passive prevents OSPF neighbor formation and stops routing updates on that link, which reduces unnecessary OSPF traffic. The network is still advertised to OSPF neighbors, so reachability information remains available.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF uses a hierarchical area structure with Area 0 as the mandatory backbone. This confines Link-State Advertisements (LSAs) and SPF calculations to individual areas, while Area Border Routers (ABRs) summarize routes between areas. The result is much smaller link-state databases and routing tables, allowing OSPF to scale to very large networks.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

interfaces in the range 10.0.0.0 to 10.255.255.255? `network 10.0.0.0 0.255.255.255 area 0`

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used. Cisco routers prefer the path with the lowest Administrative Distance first. EIGRP has AD 90, OSPF has AD 110, and RIP has AD 120, so EIGRP wins regardless of the actual metric values.